SMITHIAN FOLD THEORY V3

From Nothing to Fold

A premise-free, parameter-free and machine-closed Foundation for Smithian Fold Theory

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> ***Publication control.*** *This is a new versioned successor. It does not rewrite the preceding paper, its DOI record, historical receipts or evidence. No push, upload, DOI action, release or publication is authorised until Maria Smith explicitly confirms the exact final files and hashes.*
>
> ***Lean verification integration.*** *The current SFT ordered census passed the independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims, 898,902 candidate records and decisions, 11,108 controls, seventeen branches and no reported issue. Lean natively proves the two-class operational root and its unique survivor and constructs proof-bearing acceptance-gate certificates. The remaining claim content is checked as the complete registered artifact graph; that distinction is retained throughout this successor.*

Abstract

This successor presents the complete current `foundation` paper surface: 16 live model-admitted claims, 5,222 generated candidates, 16 unique survivors and 64 passed controls. It integrates 0 claim section(s) not present in the preceding abstract's dated surface and remains complete only to this versioned registered census, with lawful extension open. The independent Lean 4 layer returned PASS for this surface as part of all 2,777 current claims.

The patch adds six laws absent from version 1.0: the complete exact operational interface; the uniquely forced half-One ground; the exact two-preimage Fold dynamics and uniform-partition theorem; the four-map and 84-composition least-size uniqueness theorem; the depth-independent replayable derivation-trace theorem; and the 2,048-path fail-closed admission theorem. It also resolves a prior terminology collision by distinguishing complement from half-One phase translation, preserving both exact results without silently identifying different operations.

This is a formal Foundation branch. It owns no physical constant, dimensional quantity or observational dataset. Accordingly, physical measurement is not an applicable evidence class here. The empirically consequential result is instead the forced one-way boundary: future empirical branches must seal an exact prediction before authoritative target data open, retain every row and preserve both accepted and rejected receipts. The paper includes the complete dependency, candidate, elimination, control, certificate and receipt route for every theorem.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an

empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned `PASS` for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.
5. **Current paper surface.** This successor accounts for all 16 current claim(s) in its declared branch ownership boundary; 0 later claim section(s) are integrated in the successor supplement.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

Current status, evidence language and reader map

Status field	Current position
Paper and completion boundary	16 current claims in the declared branch surface. Completion is dated to the current registered census and remains open to lawful versioned extension.
Formal status	The Lean root theorem is kernel checked. Every live model claim has a current admitted receipt and passed the Lean artifact gates. Formal admission does not imply empirical confirmation.
Empirical status	Claim specific. Blind, non-blind, development-observed, holdout, favourable, adverse, null, missing, unavailable, disputed and unresolved distinctions remain exactly as recorded.
Chronology	Claim-specific registration, sealing, custody and observation order controls. Later evidence never becomes an earlier prediction without the registered chronology.
Publication status	Version 1.4.0 is a final local publication candidate awaiting Maria Smith's approval. It is not deposited or published.
Ownership	This paper retains its declared scientific boundary. Lean verification creates no new branch ownership and no application may select a law.
What is not claimed	Permanent closure of science, universal empirical confirmation, conversion of compatibility into confirmation, or superiority to every rival model.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the named formal, computational or empirical test; its kind must be stated.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications; none substitutes for another.
Foundational closure	Completion of the registered foundation only; not field-wide closure.
Field-wide closure	Completion of the named frozen or registered dated field census.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may enter a later version without rewriting receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relations; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. They remain authoritative where prose abbreviates display.

Editorial change control

This successor improves currency, expression, typography and navigation. It does not change scientific meaning, chronology, evidence class, branch ownership, claim status or machine identity. Any conflict between authoritative records must be resolved by explicit supersession or reported to Maria Smith; prose alone cannot manufacture agreement.

1. Headline findings

1. **The root is operationally premise-free.** An unpresented absence supplies no counterexample; any presented denial is an occurrence and is therefore not nothing.

2. **The One and exact positive domain are closed.** Proof quantities are exact positive parts through the One. Numerical zero, signed values, floating point, irrationals and imaginaries are outside the derivational carrier.
3. **Fold is structurally and operationally fixed.** The minimal nontrivial reversible distinction has two equal disjoint held fibres and a return to the One; operational Fold is exact self-junction followed by complete-One cast.
4. **Half-One is derived, not fitted.** One held fibre of the first Fold is the unique proper one-of-two class whose self-junction is the One. It is self-complementary, Folds to the One and is not the same operation as a half-One phase translation.
5. **The Fold fibre is exactly two.** Every image has the lower preimage `y/2` and its half-One phase partner; both Fold to the same image. Even uniform divisions remain uniform after duplicate images are identified.
6. **Primitive uniqueness is mechanically scoped.** Four normalised primitive maps are executed. Base-four ranking produces 4, 16 and 64 composition words. Fold alone generates at size one; positive construction-size induction prevents every later word from displacing it. The claim is conditional on the explicit grammar and generator predicate.
7. **A proof is a replayable object.** Every admitted finite derivation retains immutable sources, dependency receipts, ordered exact operations, intermediate identities and a terminal identity that independent execution must reproduce.
8. **Authority is fail-closed.** Of 2,048 generated admission paths, only one preserves root custody, zero parameters, complete enumeration, unique forcing, form closure, controls, independent validation, post-seal measurement order, receipt retention and admitted-dependency use.

Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorise. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigour.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Foundation, no institution, conventional axiom set, inherited ontology or target result is allowed to fill the premise-free starting point. The public burden is stronger than declaration: every proposed form must appear in the generated census, every rejection must remain visible, and the sole survivor must be reproducible through the unchanged admission engine.

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Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact Maria.Smith.Sftoe@gmail.com, submit through https://discord.gg/ucwGryVxGr, or inspect the public project at https://github.com/MettaMazza.

2. Exact validation scoreboard

Measure	Exact result
Prior source entries reviewed	356 V1 + 407 V2 = 763
Foundation-relevant source entries	14 V1 + 10 V2 = 24
Foundation-owned atomic obligations	32 of 32 closed
Model-admitted Foundation theorems	16
Generated candidate classes	5,222
Required adverse controls	64
Implementation-distinct reproductions	16
Closure scope	16 depth-independent certificates
Axioms	none
Free, fitted or learned parameters	none
Physical measurements owned by this branch	none; not an applicable evidence class
Open Foundation obligations	0

3. Evidence constitution

Every theorem uses three separated roles. **WHY** states the scientific need and cannot force a result. **DERIVATION** receives only the root theorem, admitted V3 dependencies and registered non-answering structural inputs. It generates an explicit candidate grammar, decides every candidate and requires exactly one survivor. **CHECK** begins only after sealing and contains the independent implementation, hostile controls and-only for an empirical claim-the separately held measurement target.

Earlier SFT results are mandatory observations that define reconstruction obligations. They are not executable premises and cannot select the V3 grammar, eliminator or survivor. Conventional mathematics and physics are comparison languages after derivation, never replacement premises. Every source, census, decision, closure certificate, control, validator and engine receipt is bound by SHA-256 identity. Failure is retained; it is not edited into success.

The formal results in this paper are computational proofs at their registered grammar boundaries. Unit tests show that the engine implementation behaves as specified; they do not substitute for theorem closure. Independent validators rebuild the candidate space and exact witness without importing the scientific derivation module.

4. Prior-obligation reconciliation

The complete atomic ledger is [`census/foundation_prior_obligations.json`](../../../census/foundation_prior_obligations.json). Its non-Foundation exclusion list is enumerated and hash-bound, so the 739 reviewed non-owning entries are not silently absent.

Source	Reviewed	Foundation-relevant	Atomic Foundation obligations	Open
V1 theorem manifest	356	14	9	0
V2 OneFoldMaster	407	10	23	0

5. Dependency order

```
there is no nothing
  -> structural One
    -> positive finite count
      -> exact positive part -> cross-partition equivalence
      -> minimal Fold -> complete finite Fold-word assembly
        -> exact primitive operations -> half-One -> Fold dynamics
          -> primitive-map uniqueness
        -> recursive form grammar -> canonical form enforcement
          -> replayable derivation trace
          -> one-way measurement boundary
            -> fail-closed admission law
```

Every non-root registration names exactly the single root theorem and only model-admitted dependencies. The engine refuses an absent dependency receipt.

6. Premise-free operational root theorem

Claim: `SFT-ROOT-THERE-IS-NO-NOTHING`

> *No admissible operational statement, denial, record, proof object or derivational object is nothing: absence presents no counterexample, while every presented counterexample is an occurrence.*

WHY

The reconstruction needs a beginning that is not borrowed from an axiom. The claim tested here is exact: no object admitted to an operational derivation can be nothing. Admission, statement, denial, recording and checking each present an occurrence; a non-presentation supplies no counterexample.

DERIVATION

The generator partitions every purported operational counterexample by one relation: whether it is presented at the proof boundary. This generates the complete declared grammar:

1. `unpresented-absence` supplies no statement, denial, record, trace or identity and therefore supplies no counterexample object;
2. `presented-occurrence` supplies a statement, denial, record, trace or identity, and that presentation is itself an occurrence.

Only `presented-occurrence` enters execution, and entering execution prevents it from instantiating nothing. The decision does not inspect content, notation, length or depth, so the closure certificate is depth-independent inside the stated operational boundary.

The generated cardinality is a host record of the two named partition classes; it is not a foundational numerical object imported into the theorem.

CHECK

The engine executed both generated classes, retained both decisions, verified minimality and named-shape uniqueness, and passed false-premise, tampered-source, tampered-artifact and boundary controls. A standalone Python implementation that imports no SFT module regenerated the partition and rejected a deliberately tampered survivor.

The result does not claim knowledge of an unexpressed metaphysical domain. It closes the operational foundation required for all later machine-checkable SFT derivations.

Generated grammar and exhaustive decision

- Generation rule: Partition every purported operational counterexample by whether it emits an identifiable presentation at the proof boundary.
- Exact boundary: All operational statements, denials, records and proof challenges, independent of their internal length or notation.
- Expected and generated cardinality: `2`.
- Unique survivor: `presented-occurrence`.
- Completeness certificate: `sha256:15bdb6e089afecc0a45498eabf39d25a1c93e2f5de8346fe998860bde750ed9d`.

The complete candidate list is retained at `claims/SFT-ROOT-THERE-IS-NO-NOTHING/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
1	Without a presentation there is no counterexample object to admit.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All operational statements, denials, records and proof challenges, independent of their internal length or notation.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:b06285bf05b81202ccd4331e50c6d5e4f32db07d4f14aecbcb38ac8825ac7716`.
- Generality certificate: `sha256:67bee220b6f1c353142b2021a3c83640c1ed69235429155e6ebdf3bc1ee5a608`.
- Source manifest: `sha256:c3de6a8fc34d927dce243992a5a0de610f0c8701084ae08b3c9828838f75d915`.
- Independent implementation: `sha256:54ec0bbe6fecc15920c05fc3f75df6a66957c3c15a4076245ab22c36a7b766ca`.
- Independent certificate: `recorded within the external-validation identity`.
- Derivation seal: `sha256:19c2e2af082dc9777c13bafb3bc561c7fa5cb96a51905fb26cfa06552ebf998f`.
- External-validation identity: `sha256:7942e78a54a932551ad6f5a2ceb8c5fb9122dcae0f18510052d402f42c604e3b`.
- Model-admitted engine receipt: `sha256:711864171e4d3a2f2734f0c2890965bcd81a0228349538751a3c80699c27d669` at `receipts/engine/model_admitted/SFT-ROOT-THERE-IS-NO-NOTHING-711864171e4d3a2f.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject because no counterexample identity or trace exists	unpresented-absence is not admissible as a counterexample	True	sha256:afe5d206d2c2de5d7ec6459854a1a4dec09310b2b3514f9569325262c70090a7
tampered_source	reject source drift before enumeration	the altered identity differs from the registered identity	True	sha256:4c7f233ed8e6c655aaf1094f2123f8642425ea92e9b9176dc8598b4adaccbc36
tampered_artifact	reject because forcing no longer has a unique survivor	the tampered survivor set contains both named classes	True	sha256:a4a220c6532648805adcc82501fa683fc4c87572af689a0a9022abd97255fc0a
boundary	refuse the broadened claim because it supplies no checkable object	the theorem remains scoped to admissible operational objects	True	sha256:8d3a9ef27fade730c4c3484657e789118f8f2a67045e21fecc093da702fad8c0

7. The structural One

Claim: `SFT-FOUNDATION-ONE-001`

> *The unique minimal representation of the admitted root occurrence is its complete self-whole with no unforced addition; this structural identity is the One.*

WHY

The root theorem supplies an admitted occurrence. Before arithmetic, that occurrence needs an exact self-identity. The One names its complete self-whole; it is not imported from conventional arithmetic.

DERIVATION

Every representation is classified by its coverage of the root occurrence-none, proper, or complete-and independently by whether it introduces extra material. The complete generated product has six forms. Omission and proper fragmentation fail to retain the admitted identity. Every form containing extra material adds an unforced dependency. Only `exact-self-whole` is complete and addition-free.

CHECK

All six forms were executed. One survived. Minimality, named-shape uniqueness, source drift, false premise, artifact tampering and boundary controls passed. A standalone implementation regenerated the product and rejected a tampered extra survivor.

The closure is structural. Arithmetic, fractions, domain and Fold behaviour remain future derivations.

Generated grammar and exhaustive decision

- Generation rule: Generate the product of every root-coverage relation (none, proper, complete) with whether unforced extra material is present.
- Exact boundary: All representations classified solely by coverage of the admitted root occurrence and addition of material not supplied by that occurrence.
- Expected and generated cardinality: `6`.
- Unique survivor: `exact-self-whole`.
- Completeness certificate: `sha256:4d733e780b0e83a927e6c34268290ee82101f0a1f790e523fceb2ca3b581d892`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-ONE-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
2	It presupposes a cut and fails to retain the complete admitted identity.
2	It does not retain the admitted root occurrence.
1	It introduces material not supplied by the sole admitted dependency.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All representations classified solely by coverage of the admitted root occurrence and addition of material not supplied by that occurrence.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:87c7bd8ee1b561223c7d93309ebfd8662b89cff819f5a38b30f50dd5c48f5b27`.
- Generality certificate: `sha256:74e4c3761ec998027e1b12ed7c5f2581cd45132d11134cee15e13dffc8c64a19`.
- Source manifest: `sha256:6ae512f49a7053affbd2587a859679ebc636a039a3e1d4a22f59a14f58c3a0f3`.
- Independent implementation: `sha256:123f67f4797383feb1a659ec3b66812e0870e507a81ca0b061434bd273fad66b`.
- Independent certificate: `recorded within the external-validation identity`.
- Derivation seal: `sha256:c62bb00cc30970ecf013bbc8225547a09395cf0719d3525f2a69bca1d582da7b`.

- External-validation identity: `sha256:ceaf767421b3450a862b10a6962da027c0467773a2e1be5115f554d2c3356083`.
- Model-admitted engine receipt: `sha256:68332624c276dc5d0ddfd568a26b95e20d7d5665fcdb7e825ff1b7b19bf1d51c` at `receipts/engine/model_admitted/SFT-FOUNDATION-ONE-001-68332624c276dc5d.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject an omitted root as a representation of the admitted occurrence	omitted-root does not survive	True	sha256:0dceced6a8585727559c36b83c2f4b0d0061703175b7ab66d4a653b3be67600a
tampered_source	reject a changed source identity	tampered source identity differs	True	sha256:e4188d90cf7ccec55e447016899f3e85a391963b406e3ff5f914e99dcc3daaa3
tampered_artifact	reject addition of an unforced extra to the complete root	whole-plus-extra does not survive	True	sha256:c857b9ae91fe5c805ba7e9d7b170f22ea1c25045938c3ae5c2599c5f948a199b
boundary	refuse numerical magnitude claims not yet derived	the admitted meaning is structural self-wholeness only	True	sha256:f9b13fb3dc82eeffcfa15517eb3320854a0b96759feb5e9846c0c352df66bce9

8. Exact positive finite count

Claim: `SFT-FOUNDATION-COUNT-001`

> *Every generated nonempty finite succession of structural-One occurrences has one exact positive count representation: its complete generation trace, retaining every generated occurrence once and adding none.*

WHY

The structural One supplies exact self-wholeness, but later partitions and parts require a lawful way to say how many generated occurrences are present. Importing conventional natural-number axioms would violate the clean-room boundary. This claim therefore asks what representation is forced by a registered nonempty finite generation trace itself.

DERIVATION

A proposed representation is classified by whether it retains none, a proper collection, or all generated occurrences; whether every retained occurrence appears once or at least one is duplicated; and whether material outside the registered trace is added. The grammar generates all meaningful combinations: two no-coverage forms and four forms for each of proper and complete coverage.

No coverage does not represent the registered nonempty trace. Proper coverage omits a generated occurrence. Duplication changes the trace. Extra material is ungenerated. Only `complete-coverage__once__no-extra` retains the entire trace exactly.

The depth-independent certificate has a base and successor form. The structural One is the nonempty base. Appending one newly generated occurrence makes the prior trace proper coverage of the extension; retaining the new terminal once restores completeness. The same classification therefore holds after every finite generated extension without forming a completed infinite object.

CHECK

The declared grammar contains ten generated representation classes. Every class receives a decision and proof identity. The controls reject absent coverage, source drift, duplicated occurrences and attempts to cross the positive finite boundary. A standalone implementation independently regenerates the grammar and must reject any altered second survivor.

This result establishes count identity only. Equal partitions, exact parts and Fold operations remain separate downstream obligations.

Generated grammar and exhaustive decision

- Generation rule: For no retained occurrence, generate the two ungenerated-extra classes. For proper or complete retained coverage, generate the product of exact-once or duplicated multiplicity with absence or presence of ungenerated extra occurrences.
- Exact boundary: All representations of any registered nonempty finite generation trace, classified by coverage of its generated occurrences, multiplicity of retained occurrences, and addition of occurrences outside that trace.
- Expected and generated cardinality: `10`.
- Unique survivor: `complete-coverage__once__no-extra`.
- Completeness certificate: `sha256:9734ed97836865cec30abc8cd0539c36839b4b08de3ff60cb510f1e700e9caa1`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-COUNT-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
4	It omits at least one generated occurrence and is not the complete trace.
2	It retains no occurrence from the registered nonempty trace.
2	It repeats a retained occurrence and therefore changes the generated trace.
1	It adds an occurrence not supplied by the registered generation trace.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All representations of any registered nonempty finite generation trace, classified by coverage of its generated occurrences, multiplicity of retained occurrences, and addition of occurrences outside that trace.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:5bdde2e748f2921bcb0d9f1f86181ec8bea9c104d9244977c40b36b8b9532f31`.
- Generality certificate: `sha256:c894a526f6830f0a4b714ce0b823d39a818b2e0ca6e79a252af3ab95deba7dfe`.
- Source manifest: `sha256:f7da3fcd5e30279e37e02e7801b2895711559c1070999aaa4383090dd0aecc44`.
- Independent implementation: `sha256:841876c99a3911d5beae33ee6146e13d86b8525bb680fd9b9f4661106ead39ef`.
- Independent certificate: `sha256:13e5e2f6cd9d1407ec9fb4d279fb51e9cebd3b9968afb4223306fc6befdc53a0`.
- Derivation seal: `sha256:d510105dd94110daf5c9be37ced428db4102979d4dbb41159bb1c14d9e474bcb`.
- External-validation identity: `sha256:b38c3e538cd5fd25d8628b0560ce23c9e75c59fcc82494bfe925c76884224be3`.
- Model-admitted engine receipt: `sha256:4b2411716cbef2fb9fa7aac58b24ce5eb3c0c59fe7230fd703fd89b5260f6e23` at `receipts/engine/model_admitted/SFT-FOUNDATION-COUNT-001-4b2411716cbef2fb.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject absence of a generated occurrence as a positive count	the no-coverage trace does not survive	True	sha256:c345212a50305029c7cd1b73683100c6b9364ed11d0e38c4a9eff03143a5c86b
tampered_source	reject a changed derivation source identity	the changed source identity differs	True	sha256:5e083bb4485927855fea2404ff3ce7d9bf33d6ea6247bde1e8ee0a067be5920a
tampered_artifact	reject a duplicated occurrence in the complete trace	the complete duplicated trace does not survive	True	sha256:2dae34ee14f232523153e550e31280e5e07f53ded597aab0c3045097ecba57d0
boundary	refuse zero, signed, floating or completed-infinite count claims	only nonempty finitely generated complete traces are admitted	True	sha256:c3148e0228e6336d8421753971e44b1b76ca7650c9c392cf26a561b431f1a7ca

9. Exact positive part coordinate

Claim: `SFT-FOUNDATION-PART-001`

> *For every registered finite partition of the structural One that is complete, disjoint and equal-fibred, and every nonempty held selection contained within it, the unique parameter-free exact part coordinate is the pair of positive counts identifying the held selection and the complete partition.*

WHY

Positive count alone does not identify a part of the One. A lawful part must say both how much of a generated equal partition is held and what complete partition supplies the reference. Importing division, decimal magnitude or conventional fraction axioms would select the answer from outside the clean reconstruction.

DERIVATION

The generator forms the complete product of seven criteria: complete or incomplete coverage of the One; disjoint or overlapping fibres; equal or unequal fibres; an empty, contained or outside selection; presence or absence of the held count; presence or absence of the whole count; and presence or absence of an unforced extra coordinate. This produces 192 forms.

Incomplete coverage is not the One. Overlap repeats shared structure. Unequal fibres prevent counts from identifying equal portions. An empty selection is not a positive part, and an outside selection exceeds the registered whole. The held count is necessary to identify the selection; the whole count is necessary to identify its partition reference. Any further scale is unforced. Exactly one form survives:

`complete__disjoint__equal__within__held-present__whole-present__no-extra`.

The criterion product is independent of the generated positive finite partition depth. Containment forces the held selection to remain through the complete whole, while the count dependency excludes empty, signed, floating and completed-infinite quantities.

CHECK

All 192 generated alternatives receive decisions and proof identities. Controls reject an empty selection, changed source, outside selection and any attempt to claim decimal, signed, infinite or cross-partition equivalence results. A standalone implementation independently regenerates the full product and unique survivor.

This claim deliberately leaves equivalence between differently generated held/whole coordinates open. That requires a separate common-refinement theorem before a complete rational-number structure may be claimed.

Generated grammar and exhaustive decision

- Generation rule: Generate the complete product of partition coverage, partition overlap, fibre equality, selection relation, held-count presence, whole-count presence and unforced-extra-coordinate presence.

- Exact boundary: All coordinate forms for a selection relative to a registered finite partition of the structural One, classified only by completeness, disjointness, equal-fibre status, selection containment, the two exact positive count records and unforced extra data.
- Expected and generated cardinality: `192`.
- Unique survivor: `complete__disjoint__equal__within__held-present__whole-present__no-extra`.
- Completeness certificate: `sha256:4895a8fa419c9771a061dd5438ef5d1503631057fc53bf340a5c43c8122799c5`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-PART-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
96	The partition does not retain the complete structural One.
48	Overlapping fibres count shared structure more than once.
24	Unequal fibres do not make held and whole counts an exact part coordinate.
8	The selection contains material outside the registered whole partition.
8	An empty selection supplies no positive part.
4	Without the held count the selected portion is not identified.
2	Without the whole count the selection has no exact partition reference.
1	The coordinate adds a scale or datum not supplied by the partition.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All coordinate forms for a selection relative to a registered finite partition of the structural One, classified only by completeness, disjointness, equal-fibre status, selection containment, the two exact positive count records and unforced extra data.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:e80e7008fbcdc0ec5bc2e5a6171111780d8ee1f152e64191dc9e72adf3f4cc72`.
- Generality certificate: `sha256:b660f7a3857b6e6bd5187d923a63c917ad24a94109ce395c9be1c8b6f2efca81`.
- Source manifest: `sha256:1e7375833dea5049f05826932e25994afa754dbe8c3e93d124f43b43264d6cde`.
- Independent implementation: `sha256:392fcbf89cc9a6b0c751029cc7ae3146867fef69635e5559d839c16a11cc8490`.
- Independent certificate: `sha256:844da6818e5cc4232b7d7d51ffeae22d1985484b2cea724e61692dcbda20323b`.
- Derivation seal: `sha256:5249297050c03f978c4e3211d16a4e746e8d0fc8c8ab737096e3c24bfa58429a`.
- External-validation identity: `sha256:44b76f19fae5002ed0e9d50d134962d15abde27abe4f11963e52819bf707aa27`.
- Model-admitted engine receipt: `sha256:87b40affae50458ba5492af91778fbca79a8de0bc3b951e520535093c2b7b728` at `receipts/engine/model_admitted/SFT-FOUNDATION-PART-001-87b40affae50458b.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject an empty selection as a positive part	the empty-selection coordinate does not survive	True	sha256:9862327e758bdc6709de851118d955190570c412e637ed7f5db6da19dcfd9eaa
tampered_source	reject a changed derivation source identity	the changed source identity differs	True	sha256:5e083bb4485927855fea2404ff3ce7d9bf33d6ea6247bde1e8ee0a067be5920a
tampered_artifact	reject a selection that extends outside the whole partition	the outside-selection coordinate does not survive	True	sha256:a08832bb17d902362392ea4ab7229111760e867e65c4f1791e7ebde310285c80
boundary	refuse equivalence, decimal, signed or infinite claims not derived here	the closure records exact coordinates and leaves cross-partition equivalence unresolved	True	sha256:49806f230f4da2d6c8c2983bf69e3faf5a8d2b230e294fc72d0daf96d78b36a4

10. The minimal structural Fold

Claim: `SFT-FOUNDATION-FOLD-001`

> *The unique minimal parameter-free nontrivial transformation of the structural One into a reversible distinguished partition is the first positive-count extension: two equal disjoint fibres with distinct held labels and an explicit complete return relation to the One. This structure is the Fold.*

WHY

The One, positive count and exact part coordinate establish the objects needed to ask for the first nontrivial transformation without importing a binary digit, signed pair or conventional partition rule. A Fold must distinguish while retaining the complete source whole and must do so without a chosen parameter.

DERIVATION

The generator forms the complete product of seven exhaustive classifications: complete or incomplete whole coverage; disjoint or overlapping fibres; equal or unequal fibres; identity, first count extension or later count extension; absent, same or distinct-held labels; absent or present return relation; and absence or presence of unforced extra data. This produces 288 forms.

Incomplete and overlapping forms do not retain an exact partition of the One. The unqualified One supplies no selector for unequal allocation, so inequality requires added information. Identity is trivial. The first generated successor of the One is already nontrivial, making every later extension nonminimal. Absent or identical labels fail to distinguish the fibres. Without the return relation, their complete assembly is not identified with the source One. Extra data violates parameter freedom.

Exactly one form survives:

`complete__disjoint__equal__first-extension__labels-distinct-held__return-present__no-extra`.

The first count extension contains the base One and one generated successor occurrence: the forced positive count conventionally read as two. The labels are held structural identities, never positive and negative numerical quantities.

CHECK

All 288 generated forms receive a decision and proof identity. Controls reject the unchanged identity, source drift, a later overbuilt extension and attempts to introduce signed labels, asymmetric parameters or completed-infinite arity. A standalone implementation independently regenerates the full product and the unique survivor.

This claim derives the minimal Fold only. Iterated Fold assembly, exact word grammar, observation and transformations remain separately registered work.

Generated grammar and exhaustive decision

- Generation rule: Generate the complete product of whole coverage, fibre overlap, fibre equality, positive-count extension class, label relation, return relation and unforced-extra-data presence.
- Exact boundary: All transformations of the structural One classified as identity, the first generated positive-count extension or a later extension, together with exhaustive whole-preservation, fibre, label, return and extra-data relations required for a nontrivial reversible distinction.
- Expected and generated cardinality: `288`.
- Unique survivor: `complete__disjoint__equal__first-extension__labels-distinct-held__return-present__no-extra`.
- Completeness certificate: `sha256:e63d42aeb8bdbb5b842ea2d7027ea67e585d0034da00cb3b80488fe11f50d8bc`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-FOLD-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
144	It loses part of the structural One rather than transforming the whole.
72	Overlapping fibres repeat shared structure and do not form an exact partition.
36	Unequal fibres introduce an allocation distinction not supplied by the unqualified One.
12	The identity count has no nontrivial fibre distinction.
12	A later count extension is not minimal because the first extension is already nontrivial.
4	Without labels the generated fibres are not operationally distinguishable.
4	The same label collapses the proposed distinction back to one observational class.
2	Without a retained return relation the complete fibre assembly is not identified with the One.
1	It adds a parameter or datum not supplied by the admitted dependencies.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All transformations of the structural One classified as identity, the first generated positive-count extension or a later extension, together with exhaustive whole-preservation, fibre, label, return and extra-data relations required for a nontrivial reversible distinction.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:1c1ae1d2b5052d2714e6c60a5376d3ec5fbb6b626b1e06caf5ab2513bc7c89c6`.
- Generality certificate: `sha256:ae7b6e3e23ba0b56914d84c29d50573b26f54d97c67580abe4c47e4716bc3286`.
- Source manifest: `sha256:d097df896d014d1fe39d19c3b8127a5296d9bc0777e3bda6dce84287cba5e178`.
- Independent implementation: `sha256:2186b581fca5cdfdeac8ad3cdec478cb6639fd195b3cd3c734b2c9199e0b76be`.
- Independent certificate: `sha256:551816b21e1f114ec65de0d88d4e929fe7f74c22ad71065463dfdd38d9668da3`.
- Derivation seal: `sha256:d23791605253f886d744f079e8b63c8f508e5eb580c823ff4983b511ff928421`.
- External-validation identity: `sha256:1530b38895b2de897dad0a2bea4dab18abe8b79ce54d91a9d09bd37726d5e9a9`.

- Model-admitted engine receipt: `sha256:9e99f91d15e2b68a5c24f825093686863c168595319c2083e6a5b52b0314e003` at `receipts/engine/model_admitted/SFT-FOUNDATION-FOLD-001-9e99f91d15e2b68a.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject the unchanged One as a nontrivial Fold	the identity-extension proposal does not survive	True	sha256:f46aee51f1c54c70c3b0ce43a1936c1a4d23c4d0466ab7a185f40f11c1c4a766
tampered_source	reject a changed derivation source identity	the changed source identity differs	True	sha256:5e083bb4485927855fea2404ff3ce7d9bf33d6ea6247bde1e8ee0a067be5920a
tampered_artifact	reject a later extension as the minimal Fold	the later-extension proposal does not survive	True	sha256:c8024e08141772ce43693bb845476c441ac6d103d0f3328a21099a1515a9f28b
boundary	refuse signed labels, asymmetric parameters and completed-infinite arity	the survivor uses two held labels from the first finite extension and no added datum	True	sha256:2eabd3b595538dc26114f25dd8c47b8679e4661e6a45167aa79df25b928de951

11. Cross-partition exact part equivalence

Claim: `SFT-FOUNDATION-PART-EQUIVALENCE-001`

> *Two admitted exact positive part coordinates identify the same part exactly when their generated pair-cell common refinement is complete, disjoint and equal-fibred and their completely lifted held selections admit a one-to-one and onto pairing without additional scale data.*

WHY

An exact held/whole coordinate records its construction, but differently generated partitions can identify the same portion. The equivalence law must be derived without importing multiplication, division, fraction reduction or decimal comparison.

DERIVATION

For every generated fibre of partition A, the common-refinement generator emits one pair cell with every generated fibre of partition B. Grouping by the first label refines A; grouping by the second refines B. Each held selection is lifted to all pair cells descending from its held source fibres. The two coordinates are equivalent exactly when the lifted selected traces admit a complete one-to-one and onto pairing.

The witness grammar independently classifies refinement coverage, overlap, fibre equality, refinement of A and B, complete lifting of both selections, four pairing classes and extra data. Its complete product contains 1,024 forms. Only the complete, disjoint, equal, doubly refining, doubly lifted, bijective, addition-free witness survives.

Identity pairings give reflexivity, reversing pairings gives symmetry, and chaining unique mates gives transitivity. Uniform further refinement preserves the pairing. The construction uses nested finite generation rather than an imported multiplication law.

CHECK

All 1,024 forms receive decisions and proof identities. A generated one-of-two versus two-of-four example has a pairing; a one-of-two versus one-of-three control does not. Source, second-survivor and arithmetic-import controls are also required. A standalone implementation regenerates the grammar and result.

Generated grammar and exhaustive decision

- Generation rule: Generate the complete product of common-refinement coverage, overlap, equal-fibre status, refinement of each source partition, complete lifting of each held selection, pairing class and unforced-extra-data presence.
- Exact boundary: All equivalence witnesses between two admitted exact positive part coordinates, using only a generated pair-cell common refinement and an exact pairing of their lifted selected-fibre traces.

- Expected and generated cardinality: `1024`.
- Unique survivor: `complete__disjoint__equal__a-refined__b-refined__a-selection-complete__b-selection-complete__bijective__no-extra`.
- Completeness certificate: `sha256:f883c0c836a4cf1c40af847db35c0b56e8169680b8498b1495fb1a22de0d393d`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-PART-EQUIVALENCE-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
512	The refinement does not cover the complete source wholes.
256	Overlapping refinement cells repeat shared structure.
128	Unequal refinement cells cannot compare selected part counts exactly.
64	The witness does not refine the first partition.
32	The witness does not refine the second partition.
16	The first held selection is not completely lifted.
8	The second held selection is not completely lifted.
2	The pairing repeats at least one selected cell in the second trace.
2	The pairing leaves selected cells in the second trace unmatched.
2	No exact pairing compares the lifted selected traces.
1	The witness adds a scale or rule not supplied by refinement and count.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All equivalence witnesses between two admitted exact positive part coordinates, using only a generated pair-cell common refinement and an exact pairing of their lifted selected-fibre traces.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:d43f0d3fe230c77f14b46dae6401b8430519dd65f9cb361b87d4c5cd7edea7f8`.
- Generality certificate: `sha256:07eb204fb07063de90b279dbe677a04824b601623ee6e2de5c1375a9a26692c5`.
- Source manifest: `sha256:897dc61a7c54687dea8197c782617b188543f0502475d63c1efc7e0cc0a7d816`.
- Independent implementation: `sha256:cba85223b82f565f4ff023a5eeb3367a3067c39e0756057cccbe2018e2ef2144`.
- Independent certificate: `sha256:98098ea9483a9fd8194120f9803ba0d2cde86f70919ca073e9a28860c0bb2ea7`.
- Derivation seal: `sha256:7f1c73373d9ee83c0a9c70960c1339048bed850752cbce4338c11f066fc4ed58`.
- External-validation identity: `sha256:4f4287e678544a3460786075db63016e822c8f21dc67fcf806e4d4ad7649b553`.
- Model-admitted engine receipt: `sha256:4f7c913f240527ec64d7688bf57fe95ae508f5445786dcbbdc2e6a7094476680` at `receipts/engine/model_admitted/SFT-FOUNDATION-PART-EQUIVALENCE-001-4f7c913f240527ec.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject unequal refined selected counts	the generated unequal example has no bijection	True	sha256:7fd2448bfd5621edd9a54b8e7c941f722a2233d98aca9f443c6078c63f3b1fc3
tampered_source	reject a changed derivation source identity	the changed source identity differs	True	sha256:5e083bb4485927855fea2404ff3ce7d9bf33d6ea6247bde1e8ee0a067be5920a
tampered_artifact	accept the generated one-of-two and two-of-four refinement pairing but reject an added survivor	the equal example has a bijection and only the registered witness shape survives	True	sha256:ee44dca41c7e2137ba135207468a227d576954b7a3113aed78064746edced271
boundary	refuse imported multiplication, division and decimal comparison	the witness uses nested generation and exact trace pairing only	True	sha256:deec574f5a1863b6973af416a28ac402a775867f31d8cfdaad64394f0d9aafd5

12. Complete finite Fold-word assembly

Claim: `SFT-FOUNDATION-FOLD-ASSEMBLY-001`

> *Every generated finite succession of Folds has one exact complete support: begin with the structural empty One word and at each Fold append each of the two held fibre labels once to every prior word, retaining unique word, transition and return traces.*

WHY

The minimal Fold supplies two held fibres, but not yet the law for repeated composition. Complete support must be generated without importing binary notation, exponentiation or an infinite word space.

DERIVATION

The base is the structural empty One word, not numerical zero. Each successor Fold appends `held-a` and `held-b` once to every prior word. This forces all and only the next words, one held label per Fold step. Distinct predecessors or distinct appended labels have distinct construction traces.

The 192-form representation grammar classifies trace coverage, word-length consistency, support completeness, duplication, transitions, returns and extra data. Only complete coverage with consistent unique complete support, retained transitions and returns, and no addition survives.

CHECK

The independent implementation regenerates supports through five successive Folds with generated counts 2, 4, 8, 16 and 32, while the proof of generality is the base/successor construction rather than those finite examples. Controls reject a missing one-step branch, source drift, duplicated or missing words and claims of imported binary arithmetic or completed-infinite support.

Generated grammar and exhaustive decision

- Generation rule: Generate the complete product of Fold-trace coverage, word-length consistency, support completeness, word uniqueness, transition records, return records and unforced-extra-data presence.
- Exact boundary: All representations of any generated finite Fold succession, with support formed recursively from the empty One word by appending each of the two held Fold labels at every successor step.
- Expected and generated cardinality: `192`.
- Unique survivor: `complete__consistent-length__support-complete__words-unique__transitions-present__returns-present__no-extra`.
- Completeness certificate: `sha256:2e129afd9ce31103caf388a02c30194af8e4fa45cadc1682572e2cb32c077e81`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-FOLD-ASSEMBLY-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
64	It retains no generated Fold step.
64	It omits at least one generated Fold step.
32	At least one word does not carry one held label per Fold step.
16	At least one recursively forced held-label extension is missing.
8	At least one generated word is repeated.
4	The predecessor-to-successor construction trace is missing.
2	The assembly lacks the complete return records to the One.
1	The representation adds a word, label or rule not supplied by Fold recursion.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All representations of any generated finite Fold succession, with support formed recursively from the empty One word by appending each of the two held Fold labels at every successor step.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:2b85c5826a7255c74055297535dc0cdaba0eabaca8a67192462b74140dd453d9`.
- Generality certificate: `sha256:362a45b031f9aa50e4e884f3cd92d4699e41026333ea30bd2209b7339b00d111`.
- Source manifest: `sha256:7e2afcd71b66cb0f0ada66f766be284a92966e1c84203205cc212b2f15aa8d88`.
- Independent implementation: `sha256:f40d104562c13a1c89816bf6c19d2628e6b373ca6286664177a11fe22715cd1e`.
- Independent certificate: `sha256:0c65d4f756669870eebb785c976dc09dba680c9ec6b3b3156349b4cf3e40aef6`.
- Derivation seal: `sha256:046463cefe0835f9b56a1d1c3a185d8588801e10b6530352ad955aa001a0beb5`.
- External-validation identity: `sha256:9c3ec0e13b46ee82bbb5c98f81f1c74fa0dc6db4dcc5a20ca171f0f20a57579c`.
- Model-admitted engine receipt: `sha256:8fa632be9e4048a5f2cf6749da4110e3a7b8b4d937051610dcce0583d7ea610d` at `receipts/engine/model_admitted/SFT-FOUNDATION-FOLD-ASSEMBLY-001-8fa632be9e4048a5.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject incomplete one-step support	a single held-label word omits the other forced branch	True	sha256:33c9ed6fd45b0a599e529d2783e0c4c6e27cf1ac49b040d5f9b0183958830019
tampered_source	reject changed source identity	changed identity differs	True	sha256:a100fc522c0df37fd0e0b75aa91cec82a224c73202c30cad2dc0100c34d1d0c0
tampered_artifact	reject duplicated or missing word support	generated depths one through five are unique, equal-length and complete	True	sha256:2ff11582661e71a84a1729084309f76e717a6a52d6c154e25b4d01cb92c82cf1
boundary	refuse imported binary numerals, exponentiation and completed-infinite support	support is generated only by finite held-label recursion	True	sha256:60d76f25e6d49dc4d008a943a81590eee20224dd6565d7b0bd7ba0b06fb2dead

13. Exact positive foundational operations

Claim: `SFT-FOUNDATION-EXACT-OPERATIONS-001`

> *The unique parameter-free primitive interface on exact parts of the One retains only exact positive rational parts through the One, casts complete turns to the One, defines Fold as self-junction followed by cast, permits Take only from a strictly larger part, identifies unison with the exact One relation, and retains a root-bound trace.*

WHY

The structural One, count and part coordinate require an operational law that cannot create a forbidden proof value or hide an imported arithmetic model.

DERIVATION

The complete 128-interface product classifies domain, full-turn representation, Fold composition, Take guard, unison identity, trace custody and added data. Only the exact positive domain, full-turn-as-One, self-junction-then-cast Fold, strictly guarded Take, exact One unison, root-bound trace and no-extra interface survives.

For arbitrary positive `n/d`, cast removes a complete One only while the retained magnitude exceeds the One. Fold joins an admitted part to itself, which cannot exceed two Ones, then casts. For admitted `a>b`, Take returns a strictly positive part. Coincidence is the exact identity relation One, not a numerical zero.

CHECK

The independent implementation regenerates all 128 candidates and recomputes `cast(2)=One`, `Fold(3/4)=1/2` and `Take(One,1/4)=3/4`. Controls reject reversed Take, source drift, a full turn changed to absence, and hostile zero, signed, inexact or beyond-One carriers. The proof is formal; no physical measurement applies.

Generated grammar and exhaustive decision

- Generation rule: Generate the complete product of domain, full-turn representation, Fold composition, Take guard, unison identity, trace custody and added-data class.
- Exact boundary: All primitive operational interfaces on exact positive rational parts of the structural One using only complete-One casting, self-junction and a strictly guarded positive Take.
- Expected and generated cardinality: `128`.
- Unique survivor: `domain-exact-positive-through-One__full-turn-as-One__fold-junction-then-cast__take-strictly-larger__unison-exact-One-relation__trace-root-bound__no-extra`.
- Completeness certificate: `sha256:136f73a87fc6c1bb17a02f0f87cf6d7dd59b35ebba4d881ff0f33a758ff2da00`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-EXACT-OPERATIONS-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
64	It admits absence, signed, irrational, imaginary or beyond-One proof values.
32	It turns a completed positive cycle into numerical absence.
16	Raw self-junction can leave the admitted part domain.
8	Unguarded removal can produce absence or a signed quantity.
4	It lacks the exact identity relation for coincident parts.
2	Its result is not bound to the generating operations and root trace.
1	It introduces an operation, literal or parameter not supplied by the dependencies.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All primitive operational interfaces on exact positive rational parts of the structural One using only complete-One casting, self-junction and a strictly guarded positive Take.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:ba6e2237fc5bb11032146b20e2e64d3d91c6670f885dff6584020c85663c9828`.
- Generality certificate: `sha256:01377d7c612c3bbbed0eba4c48327412bf802336c687fc96089606e9103148b4`.
- Source manifest: `sha256:4e78b00fb959659395fe37d2d4c52d9fd95924ae1ffc51d6718e2a4d66343a1b`.
- Independent implementation: `sha256:cdfd55a4d0569d79d100bbf8839c1dd1b62ffa9bc5cedeac81e60adf6f346c53`.
- Independent certificate: `sha256:9dad3820e9ac3341c67da8f9d1f9244417420d1b39dac67369c32c32a869b366`.
- Derivation seal: `sha256:771994f755b2e7ca391d15c65888108af987fde221b1d719b3d5bceb8e647339`.
- External-validation identity: `sha256:b2be9744625f8bcc0021a41886e43c579d1c3232bce1fc6e411c0d58042db882`.
- Model-admitted engine receipt: `sha256:5c0c14bcd2aa4f1df7eba1618f59ee9aa56be028aa6a8aa78fcec4d28139970d` at `receipts/engine/model_admitted/SFT-FOUNDATION-EXACT-OPERATIONS-001-5c0c14bcd2aa4f1d.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject an unguarded reversed Take	the exact carrier raises before constructing a forbidden part	True	sha256:5c6e72a2eb565ca5fac8440fcd04358e5037eb5f6eb0a49320be76fcca1af9f9
tampered_source	reject changed source identity	the changed identity differs	True	sha256:8e2f74c50aa905eb4f16b9fc9f5925508828336c4c178585b31eefc732a7ec52
tampered_artifact	reject a full turn represented as absence	casting two Ones returns the One	True	sha256:b7677e88a7394b4d09228635dea030200138ecf107dba2f5ff13ed8873e2d8fc
boundary	reject zero, signed, inexact and beyond-One proof carriers	all four hostile carrier constructions halt	True	sha256:aa257dc56f18168fe718d5756747d4f45563428a4d2d2aab33756cc52af7c780

14. The forced half-One ground

Claim: `SFT-FOUNDATION-HALF-ONE-001`

> *The minimal Fold forces one exact proper ground coordinate: either singleton held fibre has the part one-of-two; its self-junction is the One, its complement has the same exact coordinate, it Folds to the One, and the One is Fold-fixed. Complement and half-One phase translation remain distinct operations.*

WHY

V2 requires the ground solving `x` joined with itself to the One. V1 and V2 also used "antipode" for two different relations, so V3 must preserve both results with exact, nonconflated names.

DERIVATION

The minimal Fold contains two equal held fibres. Up to held-label exchange, its nonempty selection classes are a singleton and the whole pair. Only a singleton is proper and self-junctions to the whole. Its exact coordinate is therefore one-of-two. The other singleton is its complement and has the same coordinate. Fold of one-of-two completes the One; Fold of the One casts two complete Ones back to the One.

Complement is the retained part within the One. Phase antipode is half-One translation followed by cast. At half-One, complement retains half-One while phase translation lands at the One; the two operations are not silently identified.

CHECK

The independent implementation regenerates 128 candidate forms and recomputes the exact self-junction, self-complement, ground Fold and fixed-One identities. Controls reject whole selection, a changed coordinate, source drift and terminology conflation. No measurement is applicable to this formal structural theorem.

Generated grammar and exhaustive decision

- Generation rule: Generate the complete product of first-Fold selection class, self-junction, complement, Fold image, One fixedness, antipode terminology and added-data class.
- Exact boundary: All exact candidate grounds obtainable as nonempty selection-equivalence classes of the minimal Fold's two equal held fibres.
- Expected and generated cardinality: `128`.
- Unique survivor: `selection-singleton-equivalence-class__self-junction-One__self-complement-equivalent__fold-image-One__One-fixed__phase-antipode-distinguished__no-extra`.
- Completeness certificate: `sha256:ef81f0416a5fed76c5bc6789f9a79c756097c08e9af3fc6821d604a6aab0d10f`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-HALF-ONE-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
64	The complete two-fibre selection is the One, not the proper ground.
32	The selected class does not generate the complete first Fold.
16	It fails the equal-fibre complement symmetry.
8	It does not Fold to the complete source whole.
4	It does not retain the One as the completed Fold image.
2	It incorrectly identifies complement with a half-One phase translation.
1	It adds a selector or scale not supplied by the first Fold.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All exact candidate grounds obtainable as nonempty selection-equivalence classes of the minimal Fold's two equal held fibres.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:00038f607c64bf460ec965f250fd408fb45d163522334b95157bc9b907de4989`.
- Generality certificate: `sha256:fec01887732c5311e2c4675e157ce15b29eb5f3a2d0336fddb5fd9bea61309a1`.
- Source manifest: `sha256:1ff78102223262ef91d4604aa9dc61f21e77f24765c9de4ef8d680ee6acffe44`.

- Independent implementation: `sha256:a41d8d8387bf2488b84b9e4f8c9c40827720964707d3af14982d94dd26ea2812`.
- Independent certificate: `sha256:79e62e140138f8f7438b56492a1269b11db2fae494ca0edba9cb1238e2c9f384`.
- Derivation seal: `sha256:29905dc6d00d576900c9b4ce866e311828cc0fd8f9ec68eda23f45491bab009d`.
- External-validation identity: `sha256:43516b23806d4e0882b1efc46d9e7e7e35a4746b827b5f5e45a76a99abb72a8c`.
- Model-admitted engine receipt: `sha256:eef37a29bf86f0e880af72af355112f4f8efea97ba817ef71ef25d9e74883a1d` at `receipts/engine/model_admitted/SFT-FOUNDATION-HALF-ONE-001-eef37a29bf86f0e8.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject the One as the proper half-One ground	the whole-selection candidate does not survive	True	sha256:932f9927096ac22611767389ede7bcb475253b4893c5217016fa46da794a0870
tampered_source	reject changed source identity	changed identity differs	True	sha256:a100fc522c0df37fd0e0b75aa91cec82a224c73202c30cad2dc0100c34d1d0c0
tampered_artifact	reject a changed half-One coordinate	only one-of-two self-junctions exactly to the One	True	sha256:b72673264e9e945bbd28821fe1b7f6934b9663fae2e3ebfc2c935fd8a676b2fe
boundary	keep complement and phase antipode distinct	half-One complements to half-One but its half-turn translation casts to the One	True	sha256:39d4091d4cfbbffd6949ec1021c109772dd7b6f36711ff79b61a3845117e7353

15. Exact Fold dynamics and fibre law

Claim: `SFT-FOUNDATION-FOLD-DYNAMICS-001`

> *Exact Fold is closed on every admitted part, has exactly two preimages related by half-One phase translation, identifies each phase-antipodal pair, preserves uniformity of every generated even division after duplicate images are identified, and its first nonstatic recurrent orbit is the two-cycle one-of-three to two-of-three.*

WHY

The operational Fold must recover the V1 fibre, phase-antipode, uniform-division and closure laws and the V2 one-Fold equation at their exact stated strength.

DERIVATION

For image `y`, the lower preimage is `y/2`; the upper preimage is that part translated by half-One and cast. Both Fold to `y`, and every preimage lies in one class. Half-One translation twice adds one complete One, so it is involutive; doubling phase partners differs by one complete One, so Fold identifies them.

For any even generated count `2n`, the positions `i/(2n)` pair with `i+n`; their distinct Fold images are exactly the uniform `j/n` positions. Fold sends one-of-three to two-of-three and back, the first nonstatic recurrent orbit.

CHECK

The independent implementation regenerates all 128 candidates, checks exact phase and fibre identities across a hostile finite census, verifies six even partitions, and replays the two-cycle. The general certificate is algebraic in arbitrary positive numerator/denominator and arbitrary generated `n`; it does not rely on those controls as proof. No physical measurement applies.

Generated grammar and exhaustive decision

- Generation rule: Generate the complete product of Fold-domain closure, fibre cardinality, half-One phase collision, phase involution, uniform-division image, first recurrent cycle and added-data class.

- Exact boundary: All dynamical invariants of exact doubling-and-cast on the positive rational circle of the One, with half-One phase translation and generated even uniform partitions.
- Expected and generated cardinality: `128`.
- Unique survivor: `domain-closed__fibre-exactly-two__phase-collision-present__phase-involution-present__uniform-division-preserved__first-cycle-two__no-extra`.
- Completeness certificate: `sha256:666a54cd2493b579f5deccee6f7b8433d161bb4acd312fb4d6d7996c3da891dc`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-FOLD-DYNAMICS-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
64	An image leaves the exact positive domain.
32	The inverse relation does not retain both held preimages.
16	A half-One phase pair is not identified by Fold.
8	Two half-One phase translations fail to return the source.
4	An even uniform partition loses uniformity after duplicate images are identified.
2	The first nonstatic recurrent orbit does not close in two Fold moves.
1	It adds a dynamical selector not supplied by exact Fold.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All dynamical invariants of exact doubling-and-cast on the positive rational circle of the One, with half-One phase translation and generated even uniform partitions.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:6ead902affdf42ec7fb77e6585cc267074bc559499be89e6fa6edc75f4be976e`.
- Generality certificate: `sha256:957ea5e0bee2943ce460e89abb1b2475d7d94ec59919c2a6fa1920aca7a565a3`.
- Source manifest: `sha256:acd6dcb3242eccc57b8e762697dd6bf0d42a977431da610e3971e3b6ffdcd864`.
- Independent implementation: `sha256:5b943aa34f36799877f806060a893e1f00abf3b40d421b41dabc9aaff9b5e339`.
- Independent certificate: `sha256:dfa658133b558f132bb6e5b9409a7e3a44edd00eb91fddac7ab3bdac59e89573`.
- Derivation seal: `sha256:698cf0da9aa9efdc3a84bb650516aa937c001cf297252a28150757643f5a2df4`.
- External-validation identity: `sha256:2bf0fccc19c8c83362b7f229d7c8de8c9deea13844c7ec5fa38504d824b3ee2b`.
- Model-admitted engine receipt: `sha256:ffaa7c3740dd747aecfc858391cec140dd6c1b3c8ab219ca798fccfc1785b675` at `receipts/engine/model_admitted/SFT-FOUNDATION-FOLD-DYNAMICS-001-ffaa7c3740dd747a.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject a single-preimage Fold fibre	every control image has two distinct generated preimages	True	sha256:e83268614e456953e0e9f641234c40c87e29d9dbc29365636a5c8daf3aa630b5
tampered_source	reject changed source identity	changed identity differs	True	sha256:a100fc522c0df37fd0e0b75aa91cec82a224c73202c30cad2dc0100c34d1d0c0
tampered_artifact	reject a phase pair with unequal Fold images	all exact control pairs collide and the phase map is involutive	True	sha256:0a526cc0b2267d9a2b8f79edb7ce1b5ce4b7a34e96bce8de4a356a897588018c
boundary	retain exact even-partition and first-cycle scope	six even partitions remain uniform and the one-third orbit closes in two	True	sha256:34b67649fd5078b8fe9148f2b0d1b0136c5ec085ec39626fe17093e0bfb3fd9e

16. Mechanically enumerated primitive Fold-map uniqueness

Claim: `SFT-FOUNDATION-PRIMITIVE-MAP-UNIQUENESS-001`

> *Within the explicit normalized zero-parameter self-map grammar, the four primitive operational classes are identity, square, constant One and Fold; only Fold is both noninjective and nonstatically recurrent. Base-four enumeration executes all 84 ordered words through size three, while positive-size induction proves no larger composition can displace Fold as the unique least-size generator.*

WHY

V2 Steps 25 and 401 require an executable uniqueness theorem, an explicit grammar boundary and a mechanical extension beyond the primitive list.

DERIVATION

The normalised zero-parameter primitive self-map grammar has four operational classes: identity `x`, square `x*x`, constant One and `cast(x joined x)`, the Fold. Raw junction is not a total self-map; guarded removal is not total; multiplication by One normalizes to identity; closed whole terms cast to One.

Identity is static. Square strictly contracts every proper exact part. Constant One collapses. Fold is noninjective and contains the exact one-third/two-thirds recurrent orbit. Therefore Fold alone satisfies the declared generator predicate at size one.

Base-four ranking enumerates 4 size-one, 16 size-two and 64 size-three ordered words: 84 exactly. Every larger word has a larger positive construction size, so no later word can displace the already qualifying size-one Fold. The result is explicitly conditional on this grammar and predicate; it is not advertised as unrestricted expression-language uniqueness.

CHECK

The independent implementation regenerates the 84 words without importing the scientific module, confirms Fold at rank four as the sole survivor, and checks fibre collision and recurrence. Controls reject identity, shifted winners, source drift and scope expansion.

Generated grammar and exhaustive decision

- Generation rule: Base-four rank every ordered word over identity, square, constant-One and Fold at sizes one, two and three; decide least-size generation before any larger composition.
- Exact boundary: The normalized parameter-free self-map grammar generated from x and the One by product, self-junction, complete-One cast and ordered composition; the executed prefix contains every word of size one through three, and the least-size certificate excludes every later word from minimality.
- Expected and generated cardinality: `84`.
- Unique survivor: `size-1:Fold`.
- Completeness certificate: `sha256:95364355d2efc23c68d0a3c6678e20d749a13a528f475dfcbcb4315ebbe01889`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-PRIMITIVE-MAP-UNIQUENESS-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
80	A size-one generating self-map already exists, so every later composition is nonminimal regardless of its behaviour.
1	The constant map collapses every input to the fixed One and has no nonstatic recurrent orbit.
1	Square is injective and strictly contracts every proper exact part, so it has no nonstatic recurrent orbit.
1	Identity is injective and every orbit is static.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: The normalized parameter-free self-map grammar generated from x and the One by product, self-junction, complete-One cast and ordered composition; the executed prefix contains every word of size one through three, and the least-size certificate excludes every later word from minimality.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:5e77d114f690023c31906d32db2fdb5c212cc974ebaa780bd8bf9d3e44d9281a`.
- Generality certificate: `sha256:edd3b98f3098a6278df6c423597a61dfbac4a9dd7cffd1b84ad52f220cfdc0fb`.
- Source manifest: `sha256:4c1fdddcea163269ad5d32bba05086e9e61c9375135ce88fe1da9ab099ee55c2`.
- Independent implementation: `sha256:1c9f9d2377aa47311a54d1149f4566d94af059b17cb484c2b0406a044206b308`.
- Independent certificate: `sha256:d794bd240b37bff523e271c13a59208f6b18ccd56422c5250a6f69fafe1cc55b`.
- Derivation seal: `sha256:7b4ad45e5ca057659b3b8137df13f75ca83e52c14b104f3334697a0954deb414`.
- External-validation identity: `sha256:58da9e005a1d3dd8f4e731249e0adf66a0a530c179be2e99992fda709f0dc67a`.
- Model-admitted engine receipt: `sha256:8ca5e163461025f367a87f8105208ec72e4a13679a881690a3420142a9518996` at `receipts/engine/model_admitted/SFT-FOUNDATION-PRIMITIVE-MAP-UNIQUENESS-001-8ca5e163461025f3.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject identity as a generator	identity leaves one-third unchanged and is injective	True	sha256:fd2da653569af7967b51f930a30e6370304a00b168042c9179d18cdf3fd8e5ea
tampered_source	reject changed source identity	changed identity differs	True	sha256:a100fc522c0df37fd0e0b75aa91cec82a224c73202c30cad2dc0100c34d1d0c0
tampered_artifact	reject a shifted or second survivor	the 84-form prefix contains one declared survivor at base-four rank four	True	sha256:598b2c23b2eba24776f6506cc32f37123f841c7290c456a4ebf0b1cb2e3e6225
boundary	retain the explicit grammar and least-size scope	Fold is noninjective and closes the exact two-cycle while larger words remain nonminimal	True	sha256:7ad75ef148ce5118bca1ba090b0bee66372e4de16547f1fc5d942dd4c9e2329a

17. Complete foundational form grammar

Claim: `SFT-FOUNDATION-FORM-GRAMMAR-001`

The complete foundational form grammar has exactly two productions: the structural One leaf, and a finite Fold node with two recursively generated child forms on distinct held-labelled edges and an explicit return relation to the parent whole.

WHY

Fold words describe paths through repeated Folds. A complete foundational grammar must additionally derive every finite nested form, including nonuniform trees, without importing a conventional term or tree calculus.

DERIVATION

The terminal production is the structural One. The recursive production is a Fold node with exactly two previously generated child forms, one on each distinct held-labelled edge, plus the Fold return relation. Every construction trace is finite and ends in One leaves.

The generator enumerates 288 production-rule shapes across base inclusion, arity, labels, child provenance, return, termination and extra constructors. Only the base-including, exactly-two-child, distinctly labelled, generated, returning, finitely terminating and addition-free rule survives.

Structural induction proves both directions: every generated output is a finite One/Fold form, and every finite One/Fold assembly parses through these two productions.

CHECK

Controls reject a one-child Fold, source drift, same-label edges, ungenerated constructors and completed-infinite trees. The separate validator regenerates the entire rule grammar and production identities.

Generated grammar and exhaustive decision

- Generation rule: Generate the complete product of One-base inclusion, recursive child arity, edge-label relation, child provenance, return relation, finite termination and unforced-extra-production presence.
- Exact boundary: All recursive production-rule shapes for finite forms assembled only from the admitted structural One and minimal Fold.
- Expected and generated cardinality: `288`.
- Unique survivor: `base-included__two-children__labels-distinct-held__children-generated__return-present__finite-leaf-termination__no-extra`.
- Completeness certificate: `sha256:226c3a75f81ae3a5ab4326b3b6c0c51221f1e1958ef3275da5476faef712a815`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-FORM-GRAMMAR-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
144	Without the structural One leaf the recursion has no admitted terminal form.
48	One child omits a forced Fold fibre.
48	A later arity imports a nonminimal branching rule.
16	Unlabelled edges do not retain Fold distinction.
16	Identical edge labels collapse the two Fold fibres.
8	Ungenerated children introduce forms outside the dependency grammar.
4	The node lacks its complete return relation to the parent whole.
2	The production admits an uncompleted regress instead of a generated finite form.
1	The grammar adds a constructor not forced by One or Fold.

Closure and independent check

- Closure scope: `depth_independent`.

- Exact closure boundary: All recursive production-rule shapes for finite forms assembled only from the admitted structural One and minimal Fold.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:d2354df6ebb1f515a3c914c98175c20618c836434e6d3b75e803792b4ae5d331`.
- Generality certificate: `sha256:ffc1990df2db9125890fbf0d6df0a1a43b407ee203903412f386947e96052f55`.
- Source manifest: `sha256:ff7dd1129dc75d783f34d0e8878028bbcfc74eadb3f4124ce5c40e8cd62efa27`.
- Independent implementation: `sha256:2313ad9bfbd467548e5c1512adeb19283baf8178d78627b14037be77bc504818`.
- Independent certificate: `sha256:10bc7d98f5e29cbedaaf1bc003a31b11b7bda09934de1f994cbc4a63ab1c06be`.
- Derivation seal: `sha256:ff76ea58a2fa231cb763fbc16b9ddd847b51a63717c8974ae61796a27335d45d`.
- External-validation identity: `sha256:fa59cb605f36347840f33603bd2f2cc2cfaf729c56824b371eed87a821e76d81`.
- Model-admitted engine receipt: `sha256:b768689e7b061fdafa3a620befe79ddcee8bf5750cbbaf19f7b2a3156942be71` at `receipts/engine/model_admitted/SFT-FOUNDATION-FORM-GRAMMAR-001-b768689e7b061fda.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject a one-child Fold node	the one-child production is outside the grammar	True	sha256:cf5d539e03009fa3af00ff0946d3ec71304 4da5bc2e2bee752d190163b07757e
tampered_source	reject changed source identity	changed identity differs	True	sha256:a100fc522c0df37fd0e0b75aa91cec82a22 4c73202c30cad2dc0100c34d1d0c0
tampered_artifact	accept generated nonuniform finite trees and reject same-label edges	all generated examples parse and the same-label node does not	True	sha256:f13b80b2339fed04795d1133618a727b2 b9f359e868ecead459e03e26f120d32
boundary	refuse completed-infinite trees and unforced constructors	only finite traces ending in One are admitted	True	sha256:3a75d152f4cd488c7ba8d3b28da410eda0 3ecf2dcdc7e0d55c6c523ac1f467a6

18. Canonical foundational form enforcement

Claim: `SFT-FOUNDATION-FORM-ENFORCEMENT-001`

> *Every generated finite foundational form has one exact identity representation: its complete canonical recursive construction trace with each node once and every held label, child relation and Fold return preserved; two forms are equivalent exactly when these traces agree.*

WHY

A grammar generates forms, but scientific comparison also requires exact identity, minimality and named-shape equivalence without importing an external tree-isomorphism model.

DERIVATION

The One maps to its terminal construction trace. A Fold maps recursively to its Fold marker, held-a child trace, held-b child trace and return marker. This preserves every generated node and relation in the production order.

The 192-form enforcement grammar classifies coverage, node multiplicity, labels, children, returns, canonicality and extra data. Only the complete, once-retained, label-, child- and return-preserving canonical trace without additions survives. Two forms are equivalent exactly when those traces agree.

CHECK

Controls reject a partial trace, source drift and a label-preserving child swap while accepting exact reconstruction. Structural induction supplies the depth-independent certificate; the independent validator regenerates the complete enforcement grammar.

Generated grammar and exhaustive decision

- Generation rule: Generate the complete product of source-form coverage, node multiplicity, held-label preservation, child-relation preservation, return preservation, canonical-trace status and unforced-extra-data presence.
- Exact boundary: All exact-identity representations of forms admitted by the complete finite One/Fold grammar.
- Expected and generated cardinality: `192`.
- Unique survivor: `complete__node-once__labels-preserved__children-preserved__returns-preserved__trace-canonical__no-extra`.
- Completeness certificate: `sha256:8dbf1ba3093f6fbbc9fa49ee333399e52cdb0bd867941e0c63ca9990811ddc3b`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-FORM-ENFORCEMENT-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
64	It retains none of the generated source form.
64	It omits at least one generated node or relation.
32	It repeats a node already supplied by the construction trace.
16	It erases a held edge identity.
8	It changes a generated parent-child relation.
4	It omits a Fold return relation.
2	It does not follow the unique recursive production order.
1	It adds a node, relation or name not supplied by the form.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All exact-identity representations of forms admitted by the complete finite One/Fold grammar.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:126207c62074933cc3b142b541c3f7e4ddbe3c5ba507dea237fd6520ce23b82a`.
- Generality certificate: `sha256:0d2262fcfe1ece6bb5f13993f5027ffbb9e21fe1601e5ed672575f32c482eb57`.
- Source manifest: `sha256:77f0f0255530e33fff3a7caa1b8728c0df86f76ce314ec4051a834f495f74c9c`.
- Independent implementation: `sha256:84f653b58807119e31b86c1836ea537726d540d9aea00df123ef4bf25a5b78e4`.
- Independent certificate: `sha256:0fa909a6595d1990bbef5cf3cc8c952b028fef6f34635525b05d68576203eebc`.
- Derivation seal: `sha256:9a58a01ed042dbeb3e9786b5e560073c94eb1aca2032b0b9e5b8b5de0a259f79`.

- External-validation identity: `sha256:5ec6cd0ec00293eabedca0338e40b59dacebb02534941f42107115b715e35079`.
- Model-admitted engine receipt: `sha256:84d18de18b1345bf506138493aebd7808bb0bd00cc3b318040e00751929c7c8a` at `receipts/engine/model_admitted/SFT-FOUNDATION-FORM-ENFORCEMENT-001-84d18de18b1345bf.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject a partial trace as exact form identity	a trace without the held-b child is not canonical	True	sha256:5e20b36d04a2ea2840b558ab903752c95cceb38f49438b5bdb170162c7bf5e2e
tampered_source	reject changed source identity	changed identity differs	True	sha256:a100fc522c0df37fd0e0b75aa91cec82a224c73202c30cad2dc0100c34d1d0c0
tampered_artifact	reject a child swap while accepting an exact reconstruction	the exact trace matches and the label-preserving swapped trace differs	True	sha256:0d1ce333af5b449fc479f184644e7af2aa33d0420c9ed05d507a2ef037aef20d
boundary	refuse equivalence that discards held labels or finite construction identity	canonical equivalence preserves the entire recursive held-labelled trace	True	sha256:71d6a575b9efcd3fe83095d8e2ddfd31a254a002cca436f73905ecc186493bfc

19. Complete replayable derivation trace

Claim: `SFT-FOUNDATION-DERIVATION-TRACE-001`

> *Every admitted finite derivation has one exact complete record: immutable source and dependency identities followed by every registered operation in order with exact input and output identities; deterministic replay reproduces every intermediate and the terminal result, and the record traces through admitted dependencies to the root theorem.*

WHY

A statement cannot enter SFT authority merely because a program returned it. The complete ordered path, dependency custody and exact intermediates must be independently replayable to the root theorem.

DERIVATION

The 256-record product classifies source custody, dependency coverage, order, exactness, operation registration, replay, terminal identity and extra data. Only the complete source-bound ordered exact record survives.

The base is an admitted exact input with its dependency receipt. The successor appends one registered operation with ordered input hashes and exact output identity. Determinism preserves replay equality at every finite successor. The terminal output identity is the result; prose cannot replace it.

V2 Step 256 is reconstructed exactly: `Fold(one-of-three)` gives two-of-three; `Take(One,two-of-three)` gives one-of-three; the next Fold gives two-of-three. The independent direct route also gives two-of-three.

CHECK

The independent implementation regenerates all 256 records and recomputes both routes. Controls reject an unregistered operation, source drift, a changed terminal and a missing or reordered trace. The base/successor certificate is depth-independent for every generated finite derivation.

Generated grammar and exhaustive decision

- Generation rule: Generate the complete product of source custody, dependency coverage, step order, intermediate exactness, operation registration, replay equality, terminal identity and added-data class.
- Exact boundary: All finite derivation records assembled from admitted exact inputs and registered primitive operations, with an ordered hash-bound row for every successor step.
- Expected and generated cardinality: `256`.

- Unique survivor: `source-bound__dependencies-complete__order-preserved__intermediates-exact__operations-registered__replay-identical__terminal-identity-bound__no-extra`.
- Completeness certificate: `sha256:dcaa322ff8402ff4fbee3075cedec5a5630b20954d4d4c71387e4a069b77ce95`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-DERIVATION-TRACE-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
128	The executed derivation source has no immutable identity.
64	At least one admitted premise or input is omitted.
32	The successor order required for replay is absent.
16	A derivational intermediate is not an exact admitted value.
8	A step invokes an operation outside the admitted primitive set.
4	Independent execution does not reproduce every bound row.
2	The final result is not tied to the replayed terminal identity.
1	The record adds an ungenerated step or datum.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All finite derivation records assembled from admitted exact inputs and registered primitive operations, with an ordered hash-bound row for every successor step.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:741fcc8220f985f261b924731daf81bb0ac315eaa1ec4a244d6ccec0f43492ed`.
- Generality certificate: `sha256:79dc30dbe161cb2074ed3a3f51164ef91f6c387159d3182160e5057239c7b1df`.
- Source manifest: `sha256:943c820e31ecf9cb877088c66c0aa91f252b54b59180dbda07da81c8983f4e54`.
- Independent implementation: `sha256:c4d27d770fc2810c1f90799fb79a412ad7f0226f61cbaefa778ceaba7651b0b8`.
- Independent certificate: `sha256:4e5670f99a6e9c8311182b3c757413ca3b3dda3e8051f445c489764aafcc2cfa`.
- Derivation seal: `sha256:351f85f75a77dfcb4ae800256c383453ca21e6ce069fe8ec98fe0e2807cfb1c9`.
- External-validation identity: `sha256:22c5def63ec8064fe81ffb0ad16ebc286502429100d21fc5a3cd8857acb1f0d6`.
- Model-admitted engine receipt: `sha256:457414dd7e3a14e44b200b1ac329cdc9e2bb3527abe901c3255d7fb6f9233c5e` at `receipts/engine/model_admitted/SFT-FOUNDATION-DERIVATION-TRACE-001-457414dd7e3a14e4.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject an unregistered operation	replay halts on an unregistered row	True	sha256:9ee5105a6b8df1ba244ee3b1ca4bcf7578cc08a1c9dca1431d485db041fddb43
tampered_source	reject changed source identity	changed identity differs	True	sha256:a100fc522c0df37fd0e0b75aa91cec82a224c73202c30cad2dc0100c34d1d0c0
tampered_artifact	reject a changed intermediate or output	the changed terminal value disagrees with its bound identity	True	sha256:e32dfb7f142e7129743c024e21537bbb47f828d11bfeb3a5857926aeff76cc8c
boundary	replay the complete exact trace and agree with the independent direct route	both routes terminate at exact two-of-three	True	sha256:c5c734047d9b9d22e2ae9449966d260d09fd0013d481bc0574f1dbea9667d1b5

20. One-way derivation-to-measurement boundary

Claim: `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`

> *The unique admissible interface between SFT derivation and conventional measurement is one-way: an exact source-bound law seals its consequence before an identified external target is opened, measurement preserves every row, and no measured value, score or target may select or mutate the law.*

WHY

SFT must encounter conventional measured data without allowing decimals, targets, scores or observations to select its laws. The interface therefore requires its own exact closure rather than an informal methodological promise.

DERIVATION

Scientific testing requires a sealed consequence to leave a derivation. Clean forcing forbids measured answers from entering law selection. Together these requirements force one-way derivation-to-measurement flow. The law remains exact and source-bound; targets open after the seal; all favourable, unfavourable, failed and tampered rows remain; the data source is identified; measurement may validate, invalidate or bound but never mutate the law.

The generator executes 256 interface forms across flow, exactness, target order, row preservation, source identity, law mutation and extra rules. Only the complete outward, post-seal, all-row, immutable interface survives.

CHECK

Controls reject pre-seal target access, source drift, selective rows and post-measurement mutation. The independent validator regenerates the full interface grammar and registered phase order.

Generated grammar and exhaustive decision

- Generation rule: Generate the complete product of information-flow direction, proof-value exactness, target-opening order, row preservation, data-source identity, law mutation and unforced-extra-rule presence.
- Exact boundary: All interfaces between an admitted exact derivation and conventional external measurement records under the SFT empirical constitution.
- Expected and generated cardinality: `256`.
- Unique survivor: `derivation-to-measurement__proof-exact__target-after-seal__rows-complete__source-identified__law-unchanged__no-extra`.
- Completeness certificate: `sha256:6e9ae9e8067f1636e08b57220b80bf7df5318efbbb18dd8fc6501af85db7dc73`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
64	Measured records flow backward and can select the law.
64	Bidirectional flow permits target feedback into derivation.
64	A disconnected interface cannot test a derived consequence.
32	An inexact value has entered the derivational proof domain.
16	Target content is available before the derived consequence is sealed.
8	Selective retention can hide unfavourable or failed outcomes.
4	The measured record lacks an independently checkable source identity.
2	Measurement changes the already selected law.
1	The interface adds an unregistered selection or comparison rule.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All interfaces between an admitted exact derivation and conventional external measurement records under the SFT empirical constitution.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:9190d3d4fa0a037902c73d1868dc5dff8a2a2db41b695ce022998c7ca72374e5`.
- Generality certificate: `sha256:5de27755ad50fb46a7acfc1809d8037bb34da699e8f24197f4b6e637d3fed00e`.
- Source manifest: `sha256:00826bd4ce69aa55d2530c6259546fd4f58f42f4a3643d04ea3416402c21b3f7`.
- Independent implementation: `sha256:7d37793433ca53e286d36d6d6d2eeaddf94bcba287106c2d9b416e46edeb3465`.
- Independent certificate: `sha256:a754a88d8cfb04a4aa8ff6f5cdc301948075b870e6f939f5b7effce09cbf5da2`.
- Derivation seal: `sha256:cfc9a4a72a184e53495e8394fa1b4fa9918e59de02f56cba05e59353b19b9abe`.
- External-validation identity: `sha256:bad3902cca82bda05d4610560363bde0399c553972af9bf276f0bced1fb9c386`.
- Model-admitted engine receipt: `sha256:a4e409f6bb8b203de746bd0c21d3f2566e8052003219490c657249e11f6f2086` at `receipts/engine/model_admitted/SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001-a4e409f6bb8b203d.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject target access before the prediction seal	the boundary protocol rejects pre-seal target access	True	sha256:d50bf0b850a3563e28bc1fccc42c93fb443c61c5fb341e0bdb386dbe37b379b2
tampered_source	reject changed derivation source identity	changed identity differs	True	sha256:d14452d3f3787b761435c81ff8a65aecc730639e7768275e16ef8b2c3a615597
tampered_artifact	reject selective rows or post-measurement law mutation	both tampered interfaces are rejected	True	sha256:c2a9b269413a304b295638ea2f578c8280103f5047915335d951341f8e0012cd
boundary	permit conventional decimals only as post-seal measured records	proof stays exact and measurement records remain quarantined	True	sha256:7299158bec1f6e65d4397d27e5203f4c26046e2d4f7b9db771968981fd91e37f

21. Single fail-closed SFT admission law

Claim: `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001`

> *The unique admissible route from a proposed law to model authority is root- and source-bound, axiom-free and zero-parameter; it completely enumerates its declared candidate grammar, leaves one survivor, closes its form, passes all adverse controls and implementation-distinct validation, opens measurements only after sealing when empirical, preserves every disposition receipt, and permits dependency use only after model admission.*

WHY

V2 Step 4 requires enforcement to be a machine law, not an intention. Every possible bypass must be represented as a rejected admission path.

DERIVATION

Eleven exhaustive axes generate 2,048 admission paths: registration, premise/parameter status, census completeness, survivor uniqueness, form closure, adverse controls, independent validation, measurement order, receipt retention, dependency authority and added bypass. Exactly one path preserves every requirement.

Removing any gate reopens a distinct prohibited capability: hidden premise, fitted value, selected neighborhood, nonunique result, open form, untested implementation, target leakage, erased failure or premature authority. Formal claims stop after independent recomputation; empirical claims must additionally cross the one-way post-seal measurement boundary. Rejection is preserved as a receipt; only finite-complete or depth-independent closure becomes a dependency.

CHECK

The independent implementation regenerates all 2,048 paths and the sole survivor. Controls inject a free parameter, changed source, second survivor, discarded failure, pre-seal target and unclosed dependency; every altered path rejects. This theorem establishes the gate relation. Repository tests separately verify the implementation of that relation.

Generated grammar and exhaustive decision

- Generation rule: Generate the complete product of registration, premise/parameter, enumeration, forcing, closure, control, independent-validation, measurement-order, receipt-retention, dependency-authority and added-rule classes.
- Exact boundary: All admission paths from a proposed V3 derivation to model authority under the single engine, including formal-only and post-seal empirical evidence modes.
- Expected and generated cardinality: `2048`.
- Unique survivor: `registration-root-bound__no-axiom-zero-parameter__census-complete__survivor-unique__form-closed__controls-complete__validation-independent__measurement-after-seal-when-required__every-receipt-preserved__dependency-after-model-admission__no-extra`.
- Completeness certificate: `sha256:0f6757eb387bd3d7979b79cb2c679497e7c26971b5c405675c8669fb2e16325e`.

The complete candidate list is retained at `claims/SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001/candidate_census.json`. The elimination summary below counts every rejected candidate exactly once by its registered decision reason:

Rejected candidates	First decisive reason
1024	The claim lacks an exact statement, source identity or root/dependency trace.
512	An axiom, fitted value, learned value or free parameter enters the derivation.
256	The candidate neighborhood is selected rather than completely generated at its boundary.
128	Forcing leaves none or more than one survivor.
64	Minimality, named-shape uniqueness or the exact boundary remains open.
32	A false-premise, tampered-source, tampered-artifact or boundary control is absent or fails.

Rejected candidates	First decisive reason
16	The result is not recomputed by an implementation-distinct validator.
8	Target content can select or mutate the derivation.
4	An adverse or rejected receipt is lost.
2	An unclosed result is used as model authority.
1	The route adds an unregistered bypass or exception.

Closure and independent check

- Closure scope: `depth_independent`.
- Exact closure boundary: All admission paths from a proposed V3 derivation to model authority under the single engine, including formal-only and post-seal empirical evidence modes.
- Minimality: `true`.
- Named-shape uniqueness: `true`.
- Closure proof hash: `sha256:5c3b5120a9fb54af8c4b2331a3924fe62cf28dd647838fd73eb090fea69e9303`.
- Generality certificate: `sha256:f38e7cc33430e83010ab7fb397518a1a7c596bd6c6fcd14ac4582701e5ddc2cc`.
- Source manifest: `sha256:7316cbe3316bd760852bf6abc08da2ec7cb63c723b88717a6c2055b8015b1def`.
- Independent implementation: `sha256:7cde0f9abd59881aca9bec38ac1e4f3315e85bbce984429451549c2758bd6a02`.
- Independent certificate: `sha256:6a67059453fb1804324584c26f7860333b97a64db6bc332e4eb64cf36e8c9ee4`.
- Derivation seal: `sha256:8fdcead5a32387716eb8309b660af770a9ff94fc201fc086af91e3f8312e55ab`.
- External-validation identity: `sha256:649faa4f163804fee68054f0e1d06f45e17329b98e3a3bea97e2313861ea4bde`.
- Model-admitted engine receipt: `sha256:0e21ffcb217271ddfab4000901691761c2b7fd423b93d27f81ab757f0300c58c` at `receipts/engine/model_admitted/SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001-0e21ffcb217271dd.json`.

Adverse controls

Control	Expected	Observed	Pass	Receipt
false_premise	reject an axiom or free parameter	the route halts when parameter freedom is false	True	sha256:62b944027904d1c5f2056f6e9d8bc668dea529f18bf57714724483e3a4b2a2aa
tampered_source	reject a changed source identity	the changed identity differs and root custody fails	True	sha256:72019b8d60501d5b0409ee1b264215a3fb969ed5d9f26fb6d48cf53f5221284c
tampered_artifact	reject a second survivor or discarded failure	both altered routes halt	True	sha256:41d75269cfbc11cfd6ae0b6ee3d48356e64242f50266ae60494d81da4501eeab
boundary	reject pre-seal target access and unclosed dependencies	both capability violations halt while the complete path passes	True	sha256:7591c90682bc057b2b8b102186fa8b63fdf332197b1d3b2ec48694fa0108071f

22. Resolution of complement and phase-antipode terminology

The prior sources contain two exact operations under one English word. V1's positional antipode is the half-One phase translation `A(x)=cast(x+half-One)`. It is an involution and satisfies `Fold(A(x))=Fold(x)`. V2 Step 24 calls the complement within the One an antipode: `C(x)=Take(One,x)` for a proper part. The half-One is self-complementary because `C(half-One)=half-One`; its phase antipode is instead the One because `A(half-One)=cast(One)=One`.

V3 does not invalidate either prior exact calculation and does not hide the word collision. It assigns separate names, operations and receipts. This is the only way to retain both statements without allowing one symbol to denote two different maps.

23. Measurement, falsification and non-claim boundary

No theorem in this Foundation inventory outputs a physical constant, unit, material property or observed frequency. Inventing a measurement here would be category error. The applicable external evidence is implementation-distinct formal reproduction. Physical and other empirical branches must satisfy the measurement-boundary theorem derived here:

1. freeze source and candidate grammar;
2. derive and seal the exact consequence without target access;
3. open an independently identified authoritative target;
4. compare every row with full units and uncertainty where applicable;
5. run deliberately false and tampered controls;
6. preserve success, failure and rejection; and
7. prohibit the comparison from mutating the law.

The Foundation branch is falsified at its declared scope by any reproducible case in which the bound candidate generator omits a form inside its boundary, two candidates survive, the independent implementation disagrees, a required control fails, a dependency cannot trace to the root receipt, an inexact or forbidden proof value enters, or the 763-entry branch review contains an omitted Foundation-owned component. Claims outside the explicit primitive grammar, completed infinite objects, arbitrary host programs, physical constants and application performance are not claimed by this paper.

24. Empirical primacy, open knowledge and institutional accountability

Science is strongest when a result can survive an unfavourable test by a person who owes the author, institution and prevailing consensus nothing. That ideal is not identical to the machinery by which modern research is funded, selected, published and amplified. Funding provenance, review selection, publication bias, access barriers and predictive opacity are therefore legitimate objects of methodological scrutiny rather than matters outside science.

The argument is empirical. Systematic reviews have reported sponsor-associated differences in favourable efficacy results and conclusions, and evidence that commercial sponsorship can shape which questions are pursued. Studies of grant review have found limited agreement, sensitivity to panel composition and unequal outcomes for some applicant groups; a Cochrane review found insufficient evidence that grant peer review improves the quality of funded research and called for experimental study. Null and negative results remain systematically underrepresented in published literatures. UNESCO's Recommendation on Open Science identifies paywalls and high article-processing charges as sources of inequality and calls for methods, software, source code and outputs to be open for rigorous scrutiny. These findings do not prove misconduct in any particular institution or study. They do establish that institutional selection and funding are not neutral substitutes for an inspectable evidence chain.

Opaque prediction presents the same epistemic problem in computational form. A hidden model can be a useful instrument and can establish bounded predictive reliability under a properly blind test. Accuracy alone does not disclose its premises, candidate alternatives, eliminated structures, information leakage, failure boundary or derivation from source to law. NIST accordingly treats validity, reliability, accountability, transparency, explainability and interpretability as related features of trustworthy AI; reproducibility research likewise requires accessible data, code, models and workflows. SFT therefore refuses three substitutions:

1. funding success is not evidential closure;
2. professional consensus is not generated uniqueness; and
3. a black-box score is not an explicit scientific law.

This is not a rejection of expertise, peer criticism, measurement, conventional correspondence, funding or machine learning as an instrument. It rejects their use as silent answer-selectors for a purportedly fundamental law. The remedy is public and executable: register the question, expose the derivation, generate the alternatives, preserve unfavourable outcomes, bind the sources, disclose the boundary and allow anyone to reproduce acceptance or rejection.

Knowledge institutions operating through scarce grants, prestige markets, subscription access and author charges inherit incentives that are not identical to free human inquiry. Even without individual bad faith, those mechanisms help determine who can spend time asking questions, which questions appear fundable, what enters the visible record and who may read it. The loss is not limited to access to known answers; it includes questions and constructions that excluded people were never resourced or authorised to pursue.

Maria Smith developed SFT outside conventional academic credential and funding routes. That biography is not evidence that SFT is correct. It is evidence of why the work, rather than the author's institutional status, must be the unit of admission. The same standard applies to insiders and outsiders: credentials cannot rescue a failed gate, and lack of credentials cannot prevent a complete, reproducible counterexample from being heard.

Why the Foundation result is empirically consequential

The Foundation branch is formal, so it does not fabricate a physical measurement for a theorem that outputs no physical quantity. Its empirical consequence is the machinery that later natural-science claims must obey. A claim is not closed because it predicts one held-out number. It must expose the complete source-to- prediction trace, seal before target access, bind the target independently, compare every preregistered row, retain unfavourable and tampered controls and halt on leakage, ambiguity or mismatch. Blindness without transparency tests an instrument; transparency without blindness permits target selection. SFT requires both when nature is the judge.

25. Open-source rights, participation and revocable authority

The Ernos Labs Smithian Fold Theory Open Science Platform and Knowledge Tree is public, inspectable, reusable and redistributable. Maria Smith retains copyright and scientific authorship. Paper and documentation text is licensed CC BY 4.0; code is Apache-2.0. Those licenses permit lawful copying, forking, testing, modification and criticism with attribution. The Ernos Labs name is a separate conformance designation: it may be used only for work adhering to the published empirical constitution, transparent evidence requirements, fail-closed engine rules and community standards.

Scientific authority in the model is narrower than authorship or publication authority. It is an accepted, independently reproducible engine receipt at an explicit boundary. If any reviewer exposes a broken dependency, omitted candidate, second survivor, failed control, seal mismatch or empirical failure, the engine must reject the package regardless of who authored, funded, reviewed or endorsed it. Authority is inspectable and revocable by evidence.

Independent replications, invalidations, omitted candidates, counterexamples and full derivation-chain submissions are invited through Maria.Smith.Sftoe@gmail.com and https://discord.gg/ucwGryVxGr. No institutional permission is required to inspect, reproduce or challenge a public claim.

26. One-command reproduction

From the repository root on macOS, Windows or Linux with Python 3:

```
python3 tools/build_foundation_prior_obligations.py
python3 tools/build_foundation_successor_bundle.py
python3 tools/verify_publication_compliance.py --branch foundation --require-ready
```

The first command rebuilds the exact prior-obligation ledger. The second reconstructs and verifies the complete Foundation Paper 001 version 1.2 evidence bundle. The third applies the strengthened current-knowledge inventory, ownership, manuscript, rendered-paper and receipt gate. Later branches are not replayed by this branch-specific publication check. No Docker container, network service or opaque predictor is required.

27. Immutable theorem ledger

Claim	Closure	Candidates	Engine receipt
SFT-ROOT-THERE-IS-NO-NOTHING	depth_independent	2	sha256:711864171e4d3a2f2734f0c2890965bcd81a0228349538751a3c80699c27d669
SFT-FOUNDATION-ONE-001	depth_independent	6	sha256:68332624c276dc5d0ddfd568a26b95e20d7d5665fcdb7e825ff1b7b19bf1d51c
SFT-FOUNDATION-COUNT-001	depth_independent	10	sha256:4b2411716cbef2fb9fa7aac58b24ce5eb3c0c59fe7230fd703fd89b5260f6e23
SFT-FOUNDATION-PART-001	depth_independent	192	sha256:87b40affae50458ba5492af91778fbca79a8de0bc3b951e520535093c2b7b728
SFT-FOUNDATION-FOLD-001	depth_independent	288	sha256:9e99f91d15e2b68a5c24f825093686863c168595319c2083e6a5b52b0314e003
SFT-FOUNDATION-PART-EQUIVALENCE-001	depth_independent	1024	sha256:4f7c913f240527ec64d7688bf57fe95ae508f5445786dcbbdc2e6a7094476680
SFT-FOUNDATION-FOLD-ASSEMBLY-001	depth_independent	192	sha256:8fa632be9e4048a5f2cf6749da4110e3a7b8b4d937051610dcce0583d7ea610d
SFT-FOUNDATION-EXACT-OPERATIONS-001	depth_independent	128	sha256:5c0c14bcd2aa4f1df7eba1618f59ee9aa56be028aa6a8aa78fcec4d28139970d
SFT-FOUNDATION-HALF-ONE-001	depth_independent	128	sha256:eef37a29bf86f0e880af72af355112f4f8efea97ba817ef71ef25d9e74883a1d
SFT-FOUNDATION-FOLD-DYNAMICS-001	depth_independent	128	sha256:ffaa7c3740dd747aecfc858391cec140dd6c1b3c8ab219ca798fccfc1785b675
SFT-FOUNDATION-PRIMITIVE-MAP-UNIQUENESS-001	depth_independent	84	sha256:8ca5e163461025f367a87f8105208ec72e4a13679a881690a3420142a9518996
SFT-FOUNDATION-FORM-GRAMMAR-001	depth_independent	288	sha256:b768689e7b061fdafa3a620befe79ddcee8bf5750cbbaf19f7b2a3156942be71
SFT-FOUNDATION-FORM-ENFORCEMENT-001	depth_independent	192	sha256:84d18de18b1345bf506138493aebd7808bb0bd00cc3b318040e00751929c7c8a
SFT-FOUNDATION-DERIVATION-TRACE-001	depth_independent	256	sha256:457414dd7e3a14e44b200b1ac329cdc9e2bb3527abe901c3255d7fb6f9233c5e
SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001	depth_independent	256	sha256:a4e409f6bb8b203de746bd0c21d3f2566e8052003219490c657249e11f6f2086
SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001	depth_independent	2048	sha256:0e21ffcb217271ddfab4000901691761c2b7fd423b93d27f81ab757f0300c58c

28. Conclusion

The Foundation branch is reconstructed at current registered V1/V2 strength: 32 of 32 owned atomic obligations closed, 16 of 16 required theorems model-admitted, 5,222 candidate classes decided, and every required independent and adverse check preserved. This is both an exhaustive technical report and an explicit open-science position: scientific authority must rest on an inspectable chain of claims whose authority ends at their stated boundaries and whose receipts can be independently replayed or invalidated.

The next dependency-ordered branch is Mathematics. Its current-knowledge paper cannot inherit Foundation's closure as a substitute for its own 763-entry ownership review, same-strength reconstruction, exact enumerations and independent checks.

Data and code availability

All source, candidate censuses, decisions, controls, certificates and receipts are contained in the Ernos Labs Smithian Fold Theory Open Science Platform and Knowledge Tree. Foundation Paper 001 version 1.0 remains preserved in the DOI version history. This version 1.2 patch expands that same paper and does not renumber it.

Foundation and full-field reconstruction roadmap - version 1.3

PUBLISHED SAME-PAPER SUCCESSOR VERSION 1.3. The preceding version 1.2 remains permanently preserved in the Zenodo version chain. This version publishes the Foundation and full-field reconstruction roadmap without altering any admitted Foundation result.

The sixteen-law Foundation inventory remains current-evidence complete at its exact registered boundary. The continuing programme strengthens grammar reach, induction, constructor uniqueness, proof portability and adversarial authority studies; it does not replay or weaken the admitted foundation.

This amendment adds no scientific admission by prose. Every result already in the paper retains its exact receipt and evidence boundary. Every future result must pass the untouched engine independently. A branch described as complete to a dated inventory remains permanently open to lawful extension, correction, falsification and discoveries that satisfy the same public standard.

Scientific ownership

Foundation owns the premise-free root, the structural One, exact positive finite counts and parts, the Fold, foundational form generation, provenance, trace closure and fail-closed admission meaning. It does not own downstream mathematics, natural measurements or application behaviour.

Layer One - foundational reconstruction

The foundational kernel must establish:

1. the self-proven statement *There is no nothing*;
2. the One as structural wholeness;
3. exact positive finite count without importing numerical zero;
4. exact positive rational parts of the One;
5. part equivalence and finite assembly;
6. the minimal Fold and its two forced fibre labels;
7. exact foundational operations and held orientation;
8. Fold transition and recurrence;
9. primitive-map and form-grammar uniqueness;
10. canonical representation and trace enforcement;
11. the one-way boundary between derivation and measurement; and
12. fail-closed submission and independent reconstruction.

The foundational paper gate requires the complete prior-source ownership audit, the complete foundational grammar, every rejection, independent certificates, tampered controls, root traces and exact statement of all domain exclusions.

Layer Two - foundational field completion

The continuing Foundation programme covers:

- mechanical enumeration beyond any formerly size-bounded grammar;
- depth-independent induction certificates for every claimed general form;
- uniqueness and minimality of all primitive constructors;
- exact identity, emptiness, boundary, holding and composition semantics;
- provenance completeness and dependency-graph closure;
- proof-object portability across implementation languages;
- formal equivalence between readable Python evidence and independent certificates;

- adversarial studies of target leakage, receipt tampering and authority substitution;
- reviewer-facing reconstruction from the public theorem alone; and
- the future V4 handoff, without allowing V4 to author V3 retrospectively.

Publication sequence

The existing Foundation editions remain immutable records. Later versions add only newly admitted foundational results, stronger general certificates, corrections or newly discovered obligations. A later edition must preserve the original derivation chain and state why the extension was required.

Handoffs

Foundation exports exact objects, operations, traces and admissibility boundaries to Mathematics and every later branch. It imports no downstream scientific result.

References

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Complete current-census successor supplement

The authoritative current scope of this paper is 16 live model-admitted claims (`foundation` 16). Counts retained inside the preceding-version narrative describe the dated freeze at which those passages were written; this successor table and the machine census control the current total.

Every current claim identifier for this paper's branch surface already occurs in the inherited scientific body. No claim-level prose delta is required.

Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 1.3.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

This version 1.3 roadmap successor preserves the complete version 1.2 evidence and reports the completed current-knowledge Foundation of the third clean-room Smithian Fold Theory reconstruction. It updates Foundation Branch Paper 001 within the same publication and DOI version chain; it is not a separately numbered paper. The live branch contains **16 model-admitted theorems**, **5,222 generated candidate classes**, **64 required adverse-control executions**, and **16 implementation-distinct reproductions**. Its V1/V2 ownership audit reviewed all **763** registered prior entries, decomposed the 24 Foundation-relevant source entries into atomic components, assigned **32** Foundation-owned obligations and closed all **32** at the registered exact strength.

The patch adds six laws absent from version 1.0: the complete exact operational interface; the uniquely forced half-One ground; the exact two-preimage Fold dynamics and uniform-partition theorem; the four-map and 84-composition least-size uniqueness theorem; the depth-independent replayable derivation-trace theorem; and the 2,048-path fail-closed admission theorem. It also resolves a prior terminology collision by distinguishing complement from half-One phase translation, preserving both exact results without silently identifying different operations.

This is a formal Foundation branch. It owns no physical constant, dimensional quantity or observational dataset. Accordingly, physical measurement is not an applicable evidence class here. The empirically consequential result is instead the forced one-way boundary: future empirical branches must seal an exact prediction before authoritative target data open, retain every row and preserve both accepted and rejected receipts. The paper includes the complete dependency, candidate, elimination, control, certificate and receipt route for every theorem.

Scope and ownership boundary

This paper owns the `foundation` surface comprising 16 current claims. Ownership identifies responsibility for derivation and falsification; it does not prevent criticism or lawful cross-branch use. Application performance, consensus, credentials and Lean acceptance cannot transfer ownership or select a law.

Mathematical, computational and empirical constitutions

The mathematical constitution retains exact generated carriers, relations, candidate grammars, decisions, survivors and closure boundaries. The computational constitution retains executable identities, complete traces, controls, independent reconstructions and receipts. The empirical constitution retains source registration, transport, pre-seal and post-seal chronology, custody, measurement or observation, uncertainty, favourable and adverse rows, missingness, unresolved records and falsification conditions. None of these evidence classes substitutes for another.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Limitations and open frontier

The current result is formally complete only to the named dated grammar, census and artifact graph. Native Lean theoremhood is presently established for the operational root and generic gate implications; the other scientific claims are artifact checked rather than individually restated as Lean propositions. Empirical validation remains claim specific, and adverse, unavailable and unresolved evidence remains open where the current records say so. New discoveries, falsifications, stronger comparisons and native claim-level formalisations may enter only through a later version without rewriting this one.

Successor machine-identity appendix

- Preceding source: `publications/successors/foundation/FROM_NOTHING_TO_FOLD_FOUNDATION_PAPER_001_V1_3.md`
- Preceding source SHA-256: `sha256:f3387b75f402150e3d88f9216a0cb92c26449ae26d4761a028c0e7872c3c5d48`
- Lean report: `generated/lean4_validation/reports/whole_model_validation.json`
- Lean report SHA-256: `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`
- Ordered census SHA-256: `sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71`

- Execution manifest SHA-256:
  `sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41`
- Protected engine seal:
  `sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a`
- Verification-authority seal:
  `sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8`
- Publication authorised: `false`
- Remote actions performed: `false`

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned `PASS`. Its immutable report identity for this successor is `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper's registered branch surface contributes **16** of the accepted claims.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar: `unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter. Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors. `#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted gate` only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Successor conclusion

Version 1.4.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes 16 current claim(s) within `foundation` to the total dependency spine.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.